

Saint Petersburg University

**SYLLABUS
FOR AN ACADEMIC DISCIPLINE**

Введение в практические инструменты искусственного интеллекта
Introduction to Artificial Intelligence Practical Tools

Language(s) of instruction

English

ECTS workload: 6

Syllabus registration number: 068068

Saint Petersburg
2025

Part 1. Characteristics of academic studies

1.1. Goals and objectives of the academic discipline

Modern students operate in a information-rich environment and need to leverage advanced tools to enhance their learning, research, and project development efficiency. Artificial Intelligence tools have become powerful assistants in processing data, writing code, and generating innovative ideas. This course is designed to provide students with hands-on, practical skills in applying AI tools to core academic and pre-professional tasks.

The objectives of the course are to teach students how to effectively utilize AI tools for data manipulation and basic analysis, to support software development workflows, and to augment creativity and problem-solving in business contexts. Students will learn to critically assess AI-generated outputs, integrate these tools into their study processes, and understand their limitations and ethical considerations. Overall, after studying this course, students should be able to significantly accelerate and improve the quality of their academic work.

1.2. Prerequisites

No prerequisites required.

1.3. Learning outcomes

After the course the students will have knowledge and skills of:

- Using AI tools to collect, clean, and perform basic statistical analysis on datasets.
- Applying AI assistants to write, debug, and document code in Python, and to manage version control.
- Leveraging AI for brainstorming, idea validation, and initial project design in business contexts.

Compe- tency	Name of competency	Indicators of achievement
OPK-1	Able to solve professional tasks based on knowledge (at an intermediate level) of economic, organizational and managerial theory.	OPK-1.1 Evaluates the relevance and expediency of applying various management theories and concepts in order to analyze and solve management problems. OPK-1.2. Uses relevant management theories and concepts to formulate proposals for solving identified management problems.

OPK-3	Able to independently make sound organizational and managerial decisions, evaluate their operational effectiveness, social significance, and ensure their implementation in a complex (including cross-cultural) and dynamic environment	OPK-3.1 Develops a strategy of cooperation and on its basis organizes the selection of team members to achieve the goal. OPK-3.2. Plans and corrects the work of the team taking into account the interests, behavioral characteristics and opinions of its members. OPK-3.3. Resolves conflicts and contradictions in business communication on the basis of taking into account the interests of all parties. OPK-3.4. Organizes discussions on a given topic and discussion of the results of the team's work with the involvement of opponents to the developed ideas. OPK-3.5. Plans teamwork, distributes assignments and delegates authority to team members.
OPK-4	Able to manage project and process activities in the organization using modern management practices, leadership and communication skills, identify and evaluate new market opportunities, develop strategies for creating and developing innovative activities and corresponding business models	OPK-4.1 Formulates, on the basis of the problem posed, a project task and a way to solve it through the implementation of project management. OPK-4.2. Develops the concept of the project within the framework of the designated problem: formulates the goal, objectives, justifies the relevance, significance, expected results and possible areas of their application. OPK-4.3. Plans the necessary resources, including taking into account their interchangeability. OPK-4.4. Develops a project implementation plan using planning tools. OPK-4.5. Monitors the progress of the project, corrects deviations, makes additional changes to the project implementation plan, clarifies the areas of responsibility of project participants.
OPK-5	Able to generalize and critically evaluate scientific research in management and related fields, to carry out research projects	OPK-5.1 Collects and systematizes of information on the problem

PKP-6	Able to develop business plans for creating new businesses based on innovations in the context of digital transformation	PKP-6.1 Builds a flexible trajectory of business planning and transformation using the tools of continuing education, taking into account the accumulated experience of professional activity and dynamically changing requirements of the labor market
UKM-1	Able to determine the range of tasks, plan, implement his own project, including entrepreneurial, in the professional sphere.	YKM-1.1. Draws up a project implementation plan based on the target result.
UKM-2	Able to establish and maintain relationships in the social and professional sphere, taking into account legal consequences, based on intolerance to corrupt behavior and manifestations of extremism	UKM-2.1. Highlights the main factors of relationships in the professional sphere UKM-2.2. Analyzes the possibility of maintaining relationships in the social sphere. UKM-2.3. Assesses the achievability of the goals of maintaining relationships.
UKM-3	Able to use methods of obtaining and working with information in the professional sphere, taking into account modern technologies of the digital economy, artificial intelligence and data science, as well as information security	UKM-3.1 Searches for sources of information in Russian and foreign languages UKM-3.2. Uses the information and communication technologies for the search, processing and presentation of information. UKM-3.3. Compilates and corrects translation of academic and professional texts from a foreign language into the state language of the Russian Federation and from the state language of the Russian Federation into a foreign one

1.4. Description and volume of active and interactive forms of academic studies

24 hours of the course are devoted to active studies in the form of masterclasses and labs within the seminars. Lections are not provided.

Part 2. Organization, structure and content of the academic discipline

2.1. Organization of the academic discipline

The classes are delivered on a distant-learning basis. The “flipped classroom” format may be implemented in this course.

2.1.1. For core courses

Workload, volume of academic work and number of students																				
Module code	In-class workload, hours												Out of class workload, hours				Volume of active and interactive academic studies, academic hours	Workload, ECTS		
	Lectures	Seminars	Consultations	Practical training	Lab exercises	Tests	Colloquia	Mid-term examination	Final examination	Final attestation	Under guidance of instructor	With presence of instructor	Using teaching materials	Mid-term examination (out of class workload)	Final examination (out of class workload)	Final attestation (out of class workload)				
BASIC TRAJECTORY																				
Full-time mode of study																				
Semester 1	0	36	2	0	0	0	0	0	0	0	2	0	174	0	0	2	24	6		
TOTAL	0	36	2	0	0	0	0	0	0	0	2	0	174	0	0	2	24	6		

Types, forms and period of mid-term examination, final examination and final attestation						
Code of module within a discipline, training, etc.	Forms of mid-term examination		Types of final examination		Types of final attestation (only for programs with final attestation and postgraduate education programs)	
	Form	Period	Type	Period	Types	Period
BASIC TRAJECTORY						
Full-time mode of study						
Semester 1			Written test, traditional form	According to the final examination schedule		

2.2. Structure and content of the academic studies

No.	Topic / course chapter	Type of class	Academic hours
	Part 1: AI for Data Work (Beginner Level)		
1	Topic 1. Introduction to AI Assistants. Prompt engineering fundamentals. Using AI to find, summarize, and explain academic sources.	Seminars	4
2	Topic 2. Data Collection & Cleaning. Using AI to generate synthetic datasets, scrape web data (conceptually), and clean messy data.	Seminars	4
3	Topic 3. Data Analysis & Visualization. Instructing AI to perform descriptive statistics, create plots and charts, and explain the results.	Seminars	4
	Part 2: AI for Programming (Beginner Level)		
4	Topic 4. AI & the Command Line. Using AI to generate Linux/Bash commands for file management and task automation.	Seminars	4
5	Topic 5. AI & Version Control (Git). Using AI to explain Git concepts, write commit messages, and resolve merge conflicts.	Seminars	4
6	Topic 6. AI & Python Programming. Using AI to write simple functions, debug code, generate documentation, and explain algorithms.	Seminars	4
	Part 3: AI for Business & Entrepreneurship		
7	Topic 7. Ideation & Market Analysis. Using AI for brainstorming product ideas, analyzing trends, and generating SWOT analyses.	Seminars	4
8	Topic 8. Business Model & Planning. Using AI to draft value proposition canvases, business model canvases, and minimal project plans.	Seminars	4
9	Topic 9. Pitch & Content Creation. Using AI to create elevator pitches, marketing copy, and presentation outlines for a project.	Seminars	4

Part 3. Support of the academic discipline

3.1. Course support

3.1.1. Guidelines and recommendations for fulfilling the course

Methodological support for contact work includes:

- the syllabus of the discipline,
- the calendar plan of classes,
- presentations of lectures on all topics of the discipline,
- as well as other guidelines and recommendations for completing the course that are provided in the course SPbU Blackboard directory.

3.1.2. Guidelines and recommendations for self-study

Methodological support of contact work includes the syllabus of the discipline, the calendar plan of classes, presentations of lectures on all topics of the discipline, home assignments, as well as other guidelines and recommendations for self-study in the course are provided in the course SPbU Blackboard directory.

3.1.3. Assessment methods of mid-term and final examinations, assessment criteria

Control forms	Weight in the course grade	Number of points in the course grade structure
Assessment of academic progress	1,0 (100%)	100

Assessment of academic progress

Assignment	Weight in the course grade	Assignment description
Mini-Project 1: Data Work A practical project requiring students to use AI tools to find, clean, analyze, and visualize a dataset, producing a short report.	30%	In-class, open book, covers topics 1, 2, 3 <u>Grading criteria:</u> • Independence of the solution – 80% of the grade • Correctness of the solution – 20%
Mini-Project 2: Programming A practical project requiring students to use AI to write a small Python script, manage it with Git, and document the process.	30%	In-class, open book, covers topics 4, 5, 6 <u>Grading criteria:</u> • Independence of the solution – 80% of the grade • Correctness of the solution – 20%
Mini-Project 3: Business AI practical project Entrepreneurship requiring students to use AI to generate a business idea, develop a basic model, and create a pitch outline.	40%	In-class, open book, covers topics 7, 8, 9 <u>Grading criteria:</u> • Independence of the solution – 80% of the grade • Correctness of the solution – 20%
Totally:	1,0 (100%)	

Final test

Form: none

Length: none

Format: none

Final test structure: none

ECTS and traditional 5-point scale correlation

“Absolute” scale

Final result in % (assessment of academic progress + exam)	ECTS grade	5-point scale
result $\geq 90\%$	A	5 / excellent
$85\% \leq \text{result} < 90\%$	B	
$83\% \leq \text{result} < 85\%$		4 / good
$75\% \leq \text{result} < 83\%$	C	
$71\% \leq \text{result} < 75\%$	D	
$65\% \leq \text{result} < 71\%$		3 / satisfactory
$50\% \leq \text{result} < 65\%$	E	
result $< 50\%$	F	2 / unsatisfactory, course failed

Evaluation by the curve (for those who get a positive (not 2F) grade for a cohort of more than 30 students):

		Place in the rating	
5-point scale		ECTS grade	
5 / excellent	Best 25%	Best 10%	A
		Next 25%	B
4 / good	Next 50%	Next 30%	C
		Next 25%	D
3 / satisfactory	Last 25%	Last 10%	E

A schedule plan of the course containing the course progress control deadlines and assessment criteria for all tasks is published at the SPbU Blackboard system.

3.1.4. Supporting materials for academic progress assessment and final examinations (controlling and measuring materials, assessment materials)

Name of competency	Indicators of achievement	Control and measuring materials
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OPK-5 Able to generalize and critically evaluate scientific research in management and related fields, to carry out research projects	OPK-5.1 Collects and systematizes of information on the problem	Individual homework tasks aim to verify the understanding of key concepts studied within the course, as well as students' analytical skills with regard to the area of studies.
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UKM-2 Able to establish and maintain relationships in the social and professional sphere, taking into account legal consequences, based on intolerance to corrupt behavior and manifestations of extremism	UKM-2.1. Highlights the main factors of relationships in the professional sphere UKM-2.2. Analyzes the possibility of maintaining relationships in the social sphere. UKM-2.3. Assesses the achievability of the goals of maintaining relationships.	Individual homework tasks aim to verify the understanding of key concepts studied within the course, as well as students' analytical skills with regard to the area of studies.
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Mini-projects examples:

Mini-Project 1 Example (Data Work):

Task: Use an AI tool to find a publicly available dataset on a topic of your interest (e.g., climate, sports, economics). Instruct the AI to help you clean the data, calculate three descriptive statistics, and create two appropriate visualizations. Submit a brief report (max 2 pages) with your prompts, the AI's output, and your interpretation of the results.

Mini-Project 2 Example (Programming):

Task: Use an AI assistant to generate a Python script that fetches the latest 5 headlines from a news website RSS feed (using the feedparser library) and prints them to the console. Use AI to debug errors and to generate a README file explaining how to run the script. Submit your code, Git history, and the README.

Mini-Project 3 Example (Business & Entrepreneurship):

Task: Use an AI tool to brainstorm three ideas for a sustainable student-focused business. Select one idea and use the AI to help you draft a Lean Canvas for it. Finally, prompt the AI to generate a 30-second elevator pitch for your idea. Submit the canvas and the pitch.

3.1.5. Materials for assessment by students of the content and quality of the delivery of the academic discipline

A standard questionnaire is available via SPbU Blackboard.

3.2. Personnel requirements**3.2.1. Requirements for the education level and (or) qualification of full-time instructors and others allowed to teach a discipline.**

In accordance with the requirements specified in SPbU.

3.2.2. Requirements for the availability of administrative and (or) other staff

In accordance with the requirements specified in SPbU.

3.3. Physical and technical facilities**3.3.1. Requirements for classroom (rooms, seats)**

Lecture classroom with a projector; computer class, wi-fi access to the Internet.

3.3.2. Required classroom equipment including non-specialized computer equipment and software

In accordance with the requirements specified in SPbU.

3.3.3. Required specialized equipment

Unspecified.

3.3.4. Required specialized software

Firefox, Google Chrome, Yandex.Browser, Internet Explorer for access to GSOM cloud IT resources (SimBA platform for IPython notebooks, Open Web UI for large language models access).

3.3.5. List and volume of required consumable materials

In accordance with the requirements specified in SPbU.

3.4. Information resources**3.4.1. Required reading**

1. Course materials published for the students on the course Blackboard page.

3.4.2. Supplementary reading

1. William E. Shotts, The Linux Command Line: A Complete Introduction. 2019. Access: URL: <https://linuxcommand.org/tlcl.php>
2. Pro Git book, written by Scott Chacon and Ben Straub and published by Apress, <https://git-scm.com/book/en/v2>
3. PEP 8 – Style Guide for Python Code, <https://www.python.org/dev/peps/pep-0008/>
4. Documentation PostgreSQL, Chapter 2. The SQL Language, <https://www.postgresql.org/docs/13/tutorial-sql.html>
5. The Missing Semester of Your CS Education: <https://missing.csail.mit.edu/>

3.4.3. Other reading

1. Website of the M. Gorky Scientific Library of St. Petersburg State University: <http://www.library.spbu.ru/>
2. The electronic catalog of the Scientific Library named after M. Gorky St. Petersburg State University: <https://ecat.library.spbu.ru/?id=EC>
3. The electronic catalog of the branch department of the Scientific Library named after M. Gorky St. Petersburg State University in the field of management: <http://irbis.gsom.spbu.ru/>
4. The list of electronic resources available at St. Petersburg State University: <http://cufts.library.spbu.ru/CRDB/SPBGU/>
5. List of electronic resources available at GSOM St. Petersburg State University: <https://gsom.spbu.ru/gsom/library/index/>
6. The list of EBS on whose platforms Russian textbooks are available at St. Petersburg State University are presented: <http://cufts.library.spbu.ru/CRDB/SPBGU/browse?name=rures&resourcetype=8>

Part 4. Author(s) of the syllabus

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